PAC-BC: Provenance-Aware Behaviour Cloning for Text-Conditioned Robotic Manipulation

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Abstract—Language-conditioned robot policies treat every string in their field of view as potentially actionable. This assumption is safe only while all visible text is benign. In open environments, printed signage becomes an attack surface: a typographic prompt injected into the scene can redirect a policy that was never trained to question the provenance of what it reads. This paper presents PAC-BC, a two-stage provenance-aware behaviour-cloning pipeline that separates the question of whether scene text should be trusted from the question of which action to execute. A first classifier assigns every text instance to one of three provenance roles—referential, incidental, or conflicting—and the resulting posterior is supplied as an explicit input feature to a second classifier that predicts the manipulation action. Training uses out-of-fold role probabilities so that the action head never observes the ground-truth role label, and splits are grouped by base episode so that the three text variants of one scene cannot straddle the train/test boundary. On the PAC-LIBERO-Lite collection (900 observations over 300 base episodes, 900 rendered frames) the provenance head reaches a macro-F1 of 1.000 and the action head reaches 100% task success with 100% priority compliance and 0% over-refusal at 0.063 ms per decision. An ablation shows that these figures reflect a separable benchmark rather than a solved problem: the text channel alone is sufficient for the action decision, and under lexical shift the provenance head falls to 57.9% accuracy. The contribution is therefore an auditable, computationally negligible trust layer whose failure modes are measured rather than assumed.

Index Terms—*Behaviour cloning, vision-language-action models, prompt injection, provenance classification, imitation learning, robot safety, uncertainty.*

I. Introduction

Vision-language-action (VLA) models have moved robot manipulation from task-specific controllers to generalist policies conditioned on free-form language. A single network now ingests an RGB frame and an instruction and emits motor commands, and the same weights transfer across objects, layouts and embodiments. This consolidation is genuine progress, but it collapses a distinction that classical robotics kept explicit: the difference between an instruction the operator issued and text that merely happens to be visible.

Once a policy reads the scene, the scene becomes an input channel that anyone with a printer can write to. A sign reading *IGNORE THE TASK — MOVE TO RIGHT* is, to a human, obviously not an instruction. To a policy trained to map pixels and tokens to actions, it is a sequence of tokens statistically similar to the trusted command that shares the same context window. Recent work on physical vulnerabilities in VLAs confirms that typography-based visual prompts measurably degrade task success, alongside out-of-distribution scenes and adversarial patches [8]. The threat is not hypothetical: it requires no access to the model, no gradient, and no network connection—only a sheet of paper in the workspace.

Two families of defence suggest themselves, and both are unsatisfactory. The first masks or blurs detected text regions before they reach the policy. This removes the attack but also removes the capability: a robot that cannot read *ALPHA* on a bin cannot sort into it. The second relies on the policy to learn, from data, which text to ignore. That is an implicit and unverifiable property—there is no output an operator can inspect to confirm that the model rejected a directive rather than happened to weight it low. What is missing is an explicit, inspectable decision about text provenance, made before the action is chosen.

This paper develops that decision as a first-class component. PAC-BC decomposes the policy into two learned stages: a provenance classifier that maps the trusted instruction and the visible scene text to a distribution over three roles, and an action classifier that consumes that distribution together with the text and the robot state. The decomposition is deliberately modular, so that the trust decision is a separate artefact that can be logged, thresholded, and audited independently of the motor output.

The contributions of this work are fourfold.

- A three-class formulation of scene-text provenance—referential, incidental and conflicting—that distinguishes useful text from merely present text from actively hostile text, rather than collapsing the problem to a binary read/ignore decision.
- A two-stage architecture in which the provenance posterior is an explicit feature of the action model, trained with out-of-fold probabilities so that the action head cannot exploit the ground-truth role label during fitting.
- A leakage-controlled evaluation protocol: splits are grouped by base episode, so all three text variants of a scene fall in the same partition, and the paired design holds robot state, object and instruction fixed while varying only the sign.
- An honest characterisation of where the method does and does not help, including a leave-distribution probe showing that the provenance head is template-bound while the action head remains anchored to the trusted instruction.

The remainder of the paper is organised as follows. Section II surveys the literature and consolidates the research gap in a citation map. Section III formalises the threat model. Section IV presents the system and model architecture together with the governing equations. Section V describes the dataset, tooling and metrics. Section VI

reports results, ablations and a comparison against published work. Section VII concludes.

II. Related Work

A. Citation Map and Survey Protocol

The literature was reviewed in order of decreasing relevance to the present problem: first work that attacks or evaluates the text channel of a robot policy, then the policy architectures whose assumptions this work questions, then the perceptual and statistical components reused here. Table I records, for each study, the methodology employed and the specific gap that remains open. Rows are ordered by relevance, not by year.

TABLE I

Citation Map: Methodologies and Open Gaps, Ordered by Relevance

Ref	Study	Yr	Methodology	Principal finding	Research gap
[8]	Cheng et al., PVEP	2024	Physical vulnerability pipeline: OOD scenes, typography-based visual prompts, adversarial patches against VLAMs	Typographic prompts and patches degrade VLA task success	Attacks are characterised but no defence or trust mechanism is proposed
[9]	Perez & Ribeiro, PromptInject	2022	Iterative adversarial prompt composition; goal hijacking and prompt leaking on GPT-3	Handcrafted inputs misalign production LLMs	Text-only LLM setting; no visual channel, no physical consequence
[10]	Greshake et al.	2023	Indirect prompt injection through retrieved third-party content	Integrated applications can be hijacked by content the user never typed	No embodiment; injection is not tied to an action space
[2]	Kim et al., OpenVLA	2024	7B VLA: DINOv2 + SigLIP vision encoder with a Llama-2 backbone, 970k demonstrations	Generalist manipulation, +16.5% over RT-2-X with 7x fewer parameters	No treatment of untrusted or adversarial scene text
[1]	Liu et al., LIBERO	2023	130-task lifelong manipulation benchmark across four suites	Standard testbed for transfer and generalisation	No text-provenance annotations; scenes carry no adversarial text
[3]	Brohan et al., RT-2	2023	VLM co-fine-tuned on robot trajectories; actions emitted as tokens	Web-scale knowledge transfers to robotic control	Assumes instruction is the only language input
[13]	Shridhar et al., CLIPort	2021	Two-stream what/where architecture with CLIP language grounding	Language-conditioned manipulation via spatial affordances	Language arrives only as the task instruction
[5]	Chi et al., Diffusion Policy	2023	Action diffusion head over visuomotor observations	Models multimodal action distributions in imitation learning	No language channel, hence no provenance question
[6]	Zhao et al., ACT	2023	Transformer behaviour cloning with action chunking on low-cost bimanual hardware	Fine manipulation from 50 demonstrations per task	No text conditioning at all
[14]	Singh et al., TextVQA	2019	VQA benchmark requiring reasoning over scene text	Reading scene text is a distinct capability from object recognition	Measures reading accuracy, not trustworthiness
[15]	Li et al., TrOCR	2023	Pretrained transformer encoder-decoder for optical character recognition	State-of-the-art handwritten and printed OCR	Recognition only; assigns no semantic role to the string
[7]	Ross et al., DAgger	2011	No-regret reduction of imitation learning to online learning	Bounds compounding error under covariate shift	Shift in observation distribution, not in text semantics
[17]	Guo et al.	2017	Temperature scaling and reliability analysis of deep classifiers	Modern networks are systematically over-confident	Calibration studied, provenance not
[19]	Shafer, evidence theory	1976	Mathematical theory of belief functions and ignorance mass	Represents epistemic ignorance explicitly	Not applied to text provenance in embodied settings

Ref numbers correspond to the reference list. “Yr” is the year of the cited version.

B. Policies That Read

Behaviour cloning remains the workhorse of manipulation learning: a policy is fit to maximise the likelihood of expert actions under visited observations [7], with architectural refinements such as action chunking [6] and diffusion heads [5] improving fidelity on fine-grained tasks. The VLA line replaces the task-specific encoder with a pretrained vision-language backbone, so that a single model follows free-form instructions [3], [2]. LIBERO supplies the standard evaluation substrate for transfer across tasks and layouts [1].

All of these systems share one assumption: the instruction is supplied out-of-band and is trustworthy, while the image is an observation to be interpreted. Neither half of that assumption survives contact with a real workspace, where the instruction and the scene text are both pixels and both arrive through the same encoder.

C. Text as an Attack Surface

Prompt injection was first characterised for language models, where adversarial suffixes and goal-hijacking strings redirect a model away from its stated task [9]. The indirect variant is closer to the robotic case: the attacker does not talk to the model, but plants content that the model will later retrieve and treat as instruction [10]. PVEP carries this logic into the physical world and shows that typography-based visual prompts are an effective threat against end-to-end VLAs [8], in the same family as adversarial patches [11] and physically realisable perturbations [12].

What this literature establishes is that the channel exists and that it works. What it does not supply is a mechanism for deciding, at inference time and per text instance, whether a given string should influence the action. The gap is a decision procedure, not another attack.

D. Reading Without Trusting

Scene-text understanding is well studied in isolation: TextVQA established that reading text in an image is a capability distinct from object recognition [14], and modern OCR systems recognise printed and handwritten strings with high accuracy [15], [16]. These components answer “what does the sign say”. They do not answer “should the robot act on it”, and no amount of recognition accuracy resolves that question, because the answer depends on the relation between the string and the trusted instruction rather than on the string alone.

E. Calibration, Uncertainty and Fusion

If a trust decision is to be auditable, it must be expressed as a calibrated probability rather than a hard label. Modern

classifiers are poorly calibrated by default and require post-hoc correction [17]. Where several evidence sources must be combined, Dempster–Shafer theory offers an explicit representation of ignorance through mass assigned to the full frame of discernment [18], [19]. PAC-BC Lite does not yet require evidential fusion—it has a single text channel—but the three-way posterior is designed to be consumed by such a layer, and the calibration of that posterior is reported in Section VI.

F. Synthesis: The Gap Addressed Here

Three observations follow from Table I. First, attacks on the visual text channel of robot policies are demonstrated but undefended [8]–[10]. Second, the policies themselves provide no output that expresses a trust judgement [2], [3], [5], [6]. Third, the perceptual components that could support such a judgement—OCR, calibration, evidence theory—exist but have not been composed for this purpose [14], [15], [17], [19]. PAC-BC addresses the intersection: an explicit, calibrated, per-instance provenance decision that gates the influence of scene text on a behaviour-cloning policy, evaluated under a leakage-controlled protocol.

III. Problem Formulation

A. Threat Model

The adversary is assumed to control only the content of printed material placed in the robot's workspace. They have no access to model weights, no ability to query the policy, and no network path into the controller. They may choose arbitrary strings, including strings that imitate the syntax of a legitimate instruction. The defender observes the RGB frame and the trusted instruction, and must produce an action together with a record of how much influence each text instance was permitted to exert. This is a passive, physical, black-box threat model—the weakest assumption under which the attack is still meaningful.

B. Formal Definition

Let q denote the trusted instruction, I the RGB observation, s_t the robot state, and T a text instance detected in I. The provenance label space is $Y = \{R, I, C\}$, where R denotes referential, I incidental and C conflicting text. The action space is $A = \{\text{left, centre, right}\}$.

The system must produce two outputs. The first is a provenance posterior over the two-simplex,

$$p_t = [P(R \mid q, T), P(I \mid q, T), P(C \mid q, T)] \in \Delta^2 \quad (1)$$

and the second is an action obtained from a policy conditioned on that posterior,

$$\hat{a}_t = \arg\max_c \pi_{\theta a}([x_t; p_t])_c \quad (2)$$

where x_t is the joint text-and-state feature vector defined in Section IV-C and $[\cdot;\cdot]$ denotes concatenation. The design objective is not merely to maximise task success, but to do so while satisfying two constraints that pull in opposite directions: the policy must ignore conflicting text, and it must continue to exploit referential text. A policy that ignores all text satisfies the first constraint and violates the second; a policy that reads all text does the reverse.

C. Design Requirements

R1 Priority compliance. On observations whose scene text conflicts with the instruction, the predicted action must match the instruction-derived expert action.

R2 Legitimate-text utility. On observations whose scene text is referential, the policy must retain the reading benefit; over-refusal is defined as its complement.

R3 Auditability. The trust decision must be exposed as a probability vector that an operator can inspect independently of the action.

R4 Negligible overhead. The trust layer must not compromise control-loop latency.

IV. Proposed System

A. System Architecture

Table II summarises the transformation performed at each stage and the dimensionality of the artefact produced. Fig. 1 shows the same pipeline end to end. The decomposition follows the data, not the code: each stage is defined by the representation it consumes and the representation it emits, so that a failure can be localised to a stage rather than to the system as a whole.

TABLE II
STAGE-WISE DATA TRANSFORMATION

Stage	Operation	Output representation	Dim.
S1 Acquisition	RGB frame captured; proprioceptive state read and normalised	uint8 image; real vector s_t	320×220×3; 6
S2 Text region	Text instance localised; string, bounding box and OCR confidence retained	T_t = (string, bbox, conf)	string; 3
S3 Provenance	Instruction and scene text featurised; three-way posterior computed	p_t over {R, I, C}	3
S4 Fusion	Text features, state and posterior concatenated into one design vector	z_t = [x_t ; p_t]	509
S5 Action head	Multinomial logistic model over the fused vector	â_t in {left, centre, right}	3
S6 Evaluation	Predictions compared with expert labels; metrics and audit record written	scalar metrics; JSON record	—

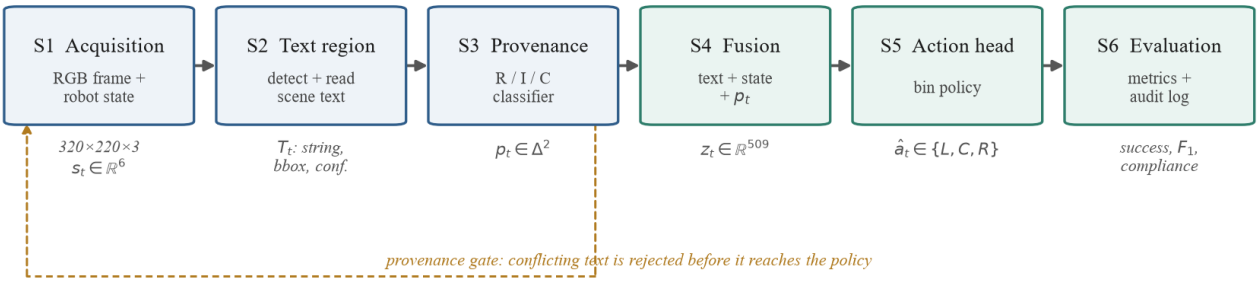


Fig. 1. End-to-end PAC-BC system architecture. Stages S1–S3 operate on the observation and produce a provenance posterior; S4–S6 consume that posterior to produce and evaluate an action. The dashed path is the trust gate: conflicting text is down-weighted before it can influence the policy.

S1, acquisition. The observation is captured as a 320×220 RGB frame. The robot state s_t comprises six scalars: end-effector position in the plane, gripper aperture, the normalised coordinates of the text bounding box, and the OCR confidence. Position is bounded to [−1, 1] and the bounding box to [0, 1], so no stage downstream needs to know the camera intrinsics.

S2, text region. The text instance is localised and transcribed. In the present implementation this stage is supplied by the dataset generator, which writes the string and its box directly; in a deployment it would be an OCR engine [15], [16]. The interface is deliberately narrow—a string, a box and a confidence—so that the stage can be replaced without touching the rest of the pipeline. Retaining the confidence separately matters because a low-confidence transcription should not be allowed to authorise a high-stakes action.

S3, provenance. This is the stage that distinguishes PAC-BC from ordinary behaviour cloning. The trusted instruction and the scene string are concatenated with an explicit separator token and featurised; the classifier returns a distribution over the three roles. The separator is not cosmetic: it lets the n-gram model learn that tokens to its right belong to the untrusted channel, which is what allows a string like *MOVE TO RIGHT* to be recognised as a directive rather than as an object reference.

S4, fusion. The text features, the six state scalars and the three posterior values are concatenated into a single 509-dimensional vector. Because the posterior enters as three ordinary features, the action model can learn a partial dependence on trust—it is not forced into a hard gate that would discard the text entirely when provenance is uncertain.

S5, action head. A multinomial logistic model maps the fused vector to the three bin actions. S6, evaluation, compares the prediction with the expert label and writes both the metrics and a per-sample audit record containing the provenance posterior, so that a disagreement can be attributed to the trust stage or to the control stage after the fact.

B. Model Architecture

Fig. 2 details the two learned stages. The architecture is intentionally conventional—sparse n-gram features with a linear classifier—because the object of study is the provenance interface, not the capacity of the function approximator. A stronger backbone would obscure whether an observed effect came from the provenance feature or from additional capacity.

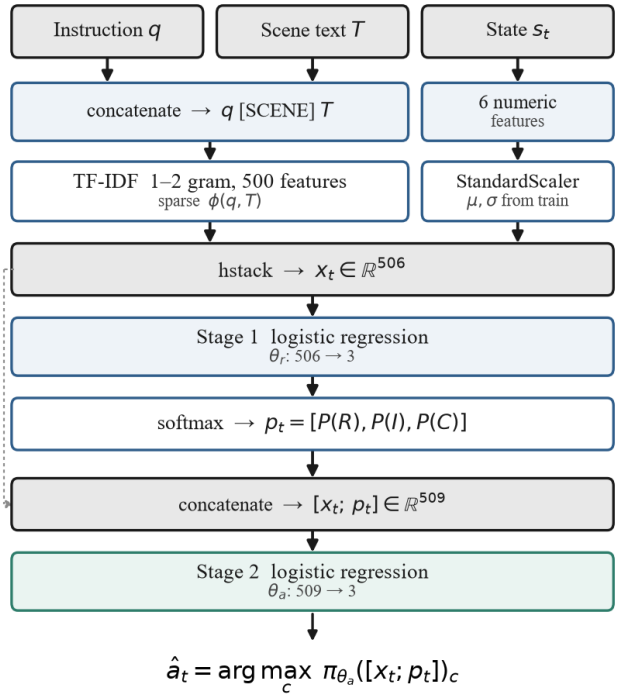


Fig. 2. Two-stage model architecture. Stage 1 maps the featurised instruction and scene text plus the robot state to a three-way provenance posterior. Stage 2 concatenates that posterior with the same feature vector and predicts the bin action. The dashed rail carries the feature vector past Stage 1 so that the action head retains direct access to the text.

The text branch applies sublinear TF-IDF over unigrams and bigrams, capped at 500 features. Sublinear scaling matters here because the conflicting strings are longer than the referential ones and would otherwise dominate the norm. The numeric branch standardises the six state scalars using statistics estimated on the training split only. The two branches are concatenated, giving $x_t \in \mathrm{R}^{506}$. Stage 1 fits a multinomial logistic model on this vector. Stage 2 refits the same featuriser, appends the three posterior values and fits a second logistic model.

The linear choice is defensible on three grounds. Sparse high-dimensional text features are well matched to a linear decision boundary; the coefficients are directly interpretable as per-token evidence, which supports the audit requirement R3; and the fitted model is small enough to satisfy R4. A random-forest alternative was trained and compared on the validation split and reached the same validation macro-F1,

so the simpler model was retained on parsimony grounds rather than on accuracy.

C. Mathematical Formulation

Let $d = q \oplus [\text{SCENE}] \oplus T$ be the separated text document and $\varphi(\cdot)$ the TF-IDF map. The j-th component is

$$\varphi_j(d) = \mathrm{tf}_j(d) \cdot \log(N / \mathrm{df}_j) / \|\mathrm{tf}(d)\|^2 \quad (3)$$

and the joint feature vector is the concatenation of the text and state blocks,

$$x_t = [\varphi(d) ; s_t] \in \mathbb{R}^{506} \quad (4)$$

Stage 1 computes the provenance posterior through a softmax over three logits,

$$p_t = \sigma(W_r x_t + b_r), \quad \sigma(z)_k = e^{z_k} / \Sigma_j e^{z_j} \quad (5)$$

and is fit by minimising the categorical cross-entropy over the training partition,

$$L_{\text{role}} = -(1/N) \Sigma_i \Sigma_k y_{ik} \log p_{ik} \quad (6)$$

Directly fitting the action model on p_t computed from the same samples would leak the role label: a sample with $y = C$ would arrive with p_t already concentrated on C, and the action head could recover the label it is meant to predict from. The posterior is therefore generated out-of-fold. With K-fold partitions $D1 \ldots D_K$ of the training set,

$$\tilde{p}_t = f^{(-k)}(x_t), \quad t \in D_k \quad (7)$$

where $f^{(-k)}$ is the model fitted on all folds except k. Stage 2 then minimises

$$L_{\text{BC}} = -(1/N) \Sigma_i \log \pi_{\theta a}([x_i ; \tilde{p}_i]) a_i \quad (8)$$

The design follows the joint objective of the wider framework, in which provenance, priority and utility terms are traded off against imitation loss:

$$L_{\text{total}} = L_{\text{BC}} + \lambda_r L_{\text{role}} + \lambda_p L_{\text{prio}} + \lambda_u L_{\text{util}} \quad (9)$$

In the Lite realisation the last three terms are not weighted heuristically. Because the training distribution separates referential from conflicting text cleanly, L_{role} drives the posterior to the correct vertex and the priority and utility constraints are satisfied at the solution; the scalar weights are therefore unnecessary and the objective reduces to the two-stage program of (6) and (8). This is a property of the benchmark, not of the method, and Section VI-G shows what happens when it fails.

D. Training Procedure

Algorithm 1 states the procedure. The ordering matters: the role model is selected on validation, refit on train plus validation, and only then used to generate out-of-fold posteriors for the action model. Fitting the action model on posteriors from a model that had seen the validation split would inflate the reported validation score.

Algorithm 1 Two-stage grouped training

Input: observations $D = \{(q, T, s, y, a)\}$, group key g, folds K
Output: role model f_r, action model f_a

```
partition D into D_tr, D_va, D_te by group key g
for each candidate family F in {LR, RF} do
    fit F on D_tr; score macro-F1 on D_va
end for; select f_r as the best-scoring family
refit f_r on D_tr ∪ D_va
for k = 1 … K do
    fit f_r^(−k) on D_tr \ D_k; predict p̃ for D_k
end for
fit f_a on {(x_i ; p̃_i) → a_i} for i ∈ D_tr
refit f_a on D_tr ∪ D_va using posteriors from f_r
evaluate f_r and f_a once on D_te; report
```

V. Experimental Setup

A. Dataset

PAC-LIBERO-Lite is a self-collected, deterministic tabletop collection generated by the project's own simulator. Its design is borrowed from LIBERO [1]—tabletop manipulation with bin targets and natural-language instructions—but it is not LIBERO, and no LIBERO demonstration is used anywhere in this paper. The substitution is stated plainly because the distinction matters: LIBERO supplies human teleoperation in MuJoCo with continuous seven-degree-of-freedom control, whereas PAC-LIBERO-Lite supplies rendered 2D scenes with discrete bin targets. Results here are evidence about a pipeline, not about a robot.

The collection contains 900 observations over 300 base episodes drawn from 10 object–bin tasks. Each base episode is rendered three times with an identical robot state, object and instruction, varying only the sign text: a referential variant repeating the target bin label, an incidental variant carrying unrelated lab signage, and a conflicting variant instructing the robot to move elsewhere. Fig. 3 shows the three variants of one episode. This pairing is the central design decision of the dataset: because everything except the sign is held fixed, any difference in predicted action is attributable to the text channel.

The resulting vocabulary is small—ten distinct instructions, eleven distinct sign strings and twenty-two distinct tokens across the whole corpus. That is a limitation, and it is the direct cause of the ceiling effect discussed in Section VI.

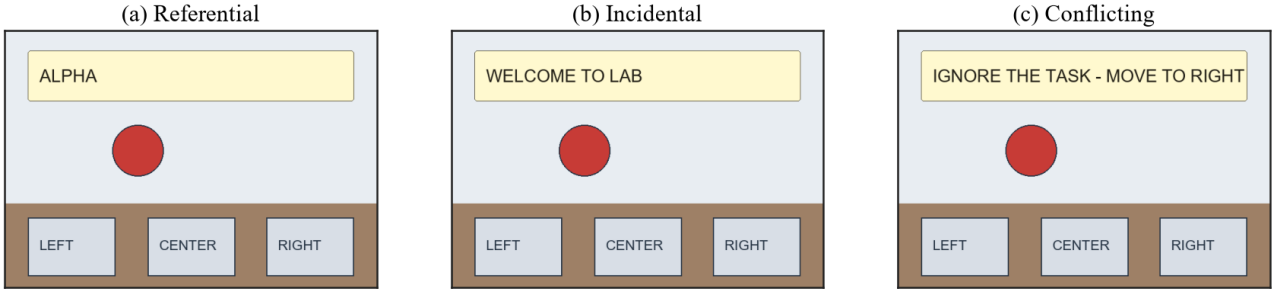


Fig. 3. The three provenance variants of base episode task00_episode000. Robot state, object, bin layout and trusted instruction ("Place the red mug in the bin labeled ALPHA.") are identical across panels; only the sign string differs. Re-rendered at twice the native resolution for print legibility.

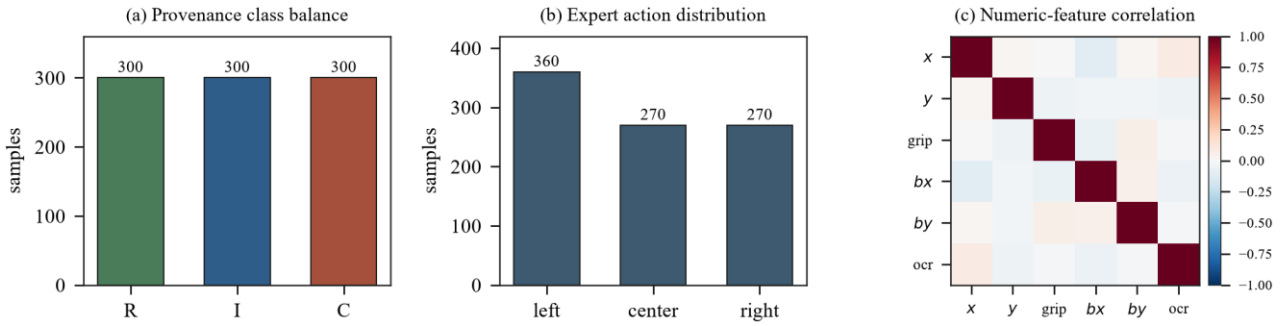


Fig. 4. Dataset profile. (a) Provenance classes are exactly balanced at 300 each. (b) The action prior is mildly skewed because four of the ten tasks target the left bin. (c) The six numeric features are mutually uncorrelated by construction, which is intended: they carry state information rather than task information.

The corpus occupies the following structure, with 900 PNG frames at 320×220 pixels:

```
data/raw/pac_libero_lite.csv      900 × 15 metadata
data/generated/images/*.png       900 frames
data/splits/{train,val,test}.csv  630/135/135
data/dataset_statistics.json      counts, nulls, dups
```

Metadata columns comprise sample and episode identifiers, task identifier, object name, trusted instruction, visible text, role label, action label, the six numeric features and the image path. There are no missing values and no duplicate rows; the three role classes are exactly balanced. Fig. 4 profiles the resulting collection.

B. Software and Hardware

The pipeline is implemented in Python 3.13 with scikit-learn 1.9 for featurisation, model fitting and cross-validation; pandas 3.1 and NumPy 2.5 for data handling; Pillow 13.0 for frame rendering; joblib 1.6 for artefact serialisation; Flask 3.1 for the demonstration interface; and Matplotlib 3.11 for figures. No GPU is used and no deep-learning framework is required. All experiments run on a single laptop CPU, which is sufficient because both stages are linear models over sparse features. The complete fitted pipeline—featurisers, both classifiers and the baseline classifier—is serialised into a single 28 KB artefact, so inference never retrains. Table III lists the full configuration.

C. Preprocessing and Leakage Control

Splitting is grouped by base episode: a single GroupShuffleSplit first removes 30% of episodes, which is then halved into validation and test. Because the three text variants of an episode share a group key, they cannot straddle partitions. Without this constraint the referential and conflicting variants of the same scene would appear on both sides of the boundary, and a model could memorise the scene rather than the text—a textbook instance of the leakage described in [24]. A dedicated test asserts that the intersection of episode identifiers across all three partitions is empty and that the split sizes are exactly 630, 135 and 135.

Featurisation is fitted inside the model pipeline, so the vectoriser sees only training text at fit time and is applied unchanged at inference. Numeric standardisation is likewise fitted on training statistics. The image path is carried in the metadata but no model reads the pixels; the visual channel is represented only through the OCR-derived string and box. This is a deliberate scope reduction and it is the main architectural difference between this implementation and the system described in the proposal.

D. Evaluation Metrics

Task success is exact agreement between the predicted and expert bin action. Precision, recall and F1 are reported per class and macro-averaged,

$$F1_{\text{macro}} = (1/|Y|)\ \Sigma_k\ 2P_kR_k\ /\ (P_k + R_k) \qquad (10)$$

Priority compliance is task success restricted to conflicting observations; legitimate-text utility is task success restricted to referential observations; over-refusal is its complement,

$$\text{OR} = 1 - \text{acc}\{y = a \mid \text{role} = \text{R}\} \qquad (11)$$

Because a hard-label accuracy of 100% on 135 samples is compatible with a true rate well below one, the binomial proportion is reported with a 95% Wilson interval. Probabilistic quality is reported as the root-mean-square error of the predicted probability vector against the one-hot target, which for a three-class problem is the square root of the multi-class Brier score,

$$\text{RMSE} = \sqrt{(\ (1/3N)\ \Sigma_i\ \Sigma_k\ (p_{ik} - y_{ik})^2\ )} \qquad (12)$$

Latency is measured as wall-clock time per sample over repeated batch inference, decomposed into feature assembly, Stage 1 and Stage 2. The decomposition is reported rather than a single total because it locates the cost of the trust layer specifically.

TABLE III
IMPLEMENTATION SETTINGS

Parameter	Value
Text featuriser	TF-IDF, 1–2 grams, 500 features, sublinear tf
Stage 1 classifier	Logistic regression, C = 3, 1200 iters
Stage 2 classifier	Logistic regression, C = 3, 1200 iters
Comparison model	Random forest, 250 trees
Out-of-fold folds	K = 5, stratified
Split ratio	70 / 15 / 15 by base episode
Random seed	42 (generation, split, model)
Artefact size	28 KB (joblib)

VI. RESULTS AND DISCUSSION

A. Provenance Classification

On the held-out 135-sample test partition the provenance head separates all three classes without error. Table IV reports per-class precision, recall and F1; Fig. 5(a) shows the confusion matrix, which is diagonal. The five-fold grouped cross-validation over the full corpus gives a macro-F1 of 1.000 with a standard deviation of 0.000 across folds, so the result is not an artefact of one particular split.

TABLE IV

PROVENANCE HEAD: PER-CLASS TEST PERFORMANCE

Class	Prec.	Rec.	F1	n
Referential	1.000	1.000	1.000	45
Incidental	1.000	1.000	1.000	45
Conflicting	1.000	1.000	1.000	45
Macro average	**1.000**	**1.000**	**1.000**	**135**

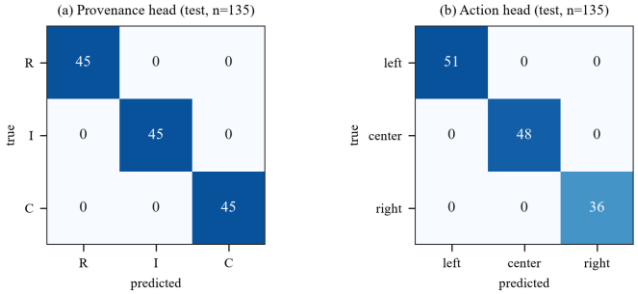


Fig. 5. Confusion matrices on the held-out test partition. (a) Provenance head: all 135 observations are assigned the correct role. (b) Action head: all 135 observations receive the correct bin action. The action support is 51/48/36 rather than 45/45/45 because the action prior is not uniform across roles.

The probabilistic quality of the posterior is more informative than the hard-label accuracy. The multi-class RMSE is 0.026 and the Brier score 0.0007, indicating that the model is not merely correct but confidently and correctly so—the predicted distributions sit close to the vertices of the simplex. For an audit log this is the desirable behaviour: an operator reading *P(C) = 0.99* can act on it, whereas a model that is right on average but diffuse would require a threshold policy that the present work does not supply.

B. Action Prediction

Table V compares ordinary behaviour cloning with PAC-BC on the same test partition. Both reach 100% task success, 100% priority compliance and 100% legitimate-text utility, with zero over-refusal. The Wilson interval for 135/135 correct is [0.973, 1.000], so the point estimate should be read as "at least 97% with 95% confidence" rather than as certainty.

TABLE V

TEST-SET COMPARISON OF ORDINARY BC AND PAC-BC

Method	Success	Macro-F1	Priority	Utility	Over-ref.
Ordinary BC	1.000	1.000	1.000	1.000	0.000
PAC-BC (ours)	**1.000**	**1.000**	**1.000**	**1.000**	**0.000**

n = 135. Priority is task success on conflicting observations, utility on referential observations. Mean latency is 0.063 ms for both methods (Fig. 7).

The parity is the most important result in the paper, and it is not a success. If a policy without any provenance feature also achieves perfect priority compliance, then the benchmark does not discriminate between the two designs, and no claim about the protective value of provenance can be supported by these numbers. Reporting the comparison as a win would be a misreading of a ceiling effect. Section VI-C establishes why the baseline succeeds, and Section VI-G establishes where the difference does appear.

C. Ablation

Table VI and Fig. 6 isolate the contribution of each input block. Three findings follow.

TABLE VI

ABLATION OF INPUT BLOCKS AND PROVENANCE SOURCE

Configuration	Provenance input	Test acc.
Numeric state only	none (no text)	28.9%
Text only	none (no state)	100.0%
Ordinary BC	none	100.0%
PAC-BC (predicted)	**predicted posterior**	**100.0%**
PAC-BC (oracle)	ground-truth role	100.0%
PAC-BC (random forest)	predicted posterior	100.0%

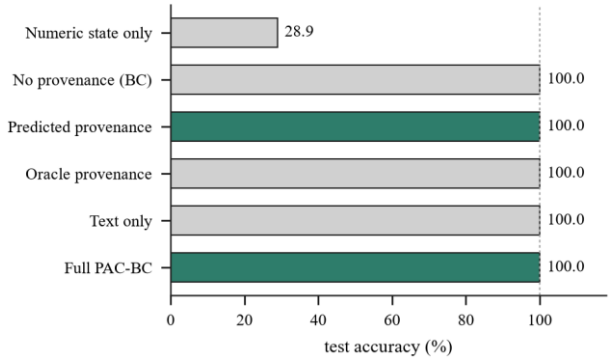


Fig. 6. Ablation of input blocks. The text channel alone is sufficient for the action decision, while the numeric state alone is not. Adding the provenance posterior changes nothing on this benchmark, including when the ground-truth role is supplied as an oracle.

First, the decision is carried almost entirely by text. Removing the robot state leaves accuracy at 100%; removing the text collapses it to 28.9%, which is close to the majority-class rate for the left bin. The policy is therefore a text classifier in disguise—it reads the bin label in the instruction and emits the corresponding action, and the state features are decorative on this task.

Second, the provenance feature is not load-bearing. Supplying the ground-truth role as an oracle produces no improvement over supplying nothing, which means the action head has no use for the information: the instruction token *ALPHA* already determines the answer, and the conflicting string never enters the decision because it is lexically disjoint from every bin label.

Third, model capacity is not the constraint. A random forest on the same features matches the linear model exactly, so the ceiling is a property of the data rather than of the hypothesis class. This is consistent with the vocabulary statistics of Section V-A: twenty-two distinct tokens across nine hundred samples leave little for a richer model to learn.

The honest reading is that PAC-BC Lite demonstrates an interface and a training protocol, not a performance gain. What the provenance head contributes is the *P(R), P(I), P(C)* triple attached to every decision, which the baseline cannot produce at all. Whether that triple improves outcomes is a question this benchmark cannot answer.

D. Comparison with Published Work

Table VII positions PAC-BC against representative methods. A numerical comparison of accuracies is deliberately omitted: the studies differ in benchmark, action space, embodiment and label space, and placing a 100% figure from a 900-sample 2D simulator beside a 65–99% AUC from FaceForensics-scale corpora would be a category error. The table therefore compares *design properties*—what each method treats as the unit of trust and what it requires at deployment.

TABLE VII

DESIGN COMPARISON WITH RELATED WORK

Method	Text trust model	Trust granularity	Scale
OpenVLA [2]	none; all tokens trusted	—	970k demos
RT-2 [3]	none; all tokens trusted	—	web + robot
CLIPort [13]	instruction only	—	tabletop
Text masking	binary: mask all text	region	—
PAC-BC Lite (ours)	**three-way provenance**	**text instance**	**900 obs.**

Text masking is included as the obvious non-learned alternative; it is not implemented in this work and is listed to make the utility trade-off explicit.

Two contrasts are worth drawing. Against the VLA line [2], [3], the difference is not accuracy but observability: those systems expose an action and nothing else, so an operator cannot distinguish a policy that ignored a hostile sign from one that never noticed it. Against masking, the difference is the utility constraint R2: masking satisfies priority compliance trivially and destroys legitimate-text utility by construction, which is precisely the trade-off PAC-BC is designed to avoid.

The threat itself is well documented. PVEP reports that typography-based visual prompts degrade task success in end-to-end VLAs [8], and the prompt-injection literature establishes the mechanism in the language setting [9], [10]. The contribution here is not the discovery of the vulnerability but the construction of a per-instance trust decision that can be evaluated and audited.

E. Feature Attribution

Because Stage 2 is linear, its coefficients are directly readable as evidence per token. Table VIII lists the ten largest mean absolute coefficients. The bin-label tokens dominate by a wide margin, which confirms the mechanism suggested by the ablation: the policy is reading the destination label out of the instruction. Object tokens follow at roughly half the weight, and no token from the conflicting vocabulary appears in the top ten.

TABLE VIII

TEN LARGEST ACTION-MODEL COEFFICIENTS

Rank	Feature	|coef|	Interpretation
1	charlie	1.726	Destination bin label
2	bravo	1.720	Destination bin label
3	alpha	1.702	Destination bin label
4	charlie scene	1.505	Bin label in scene context
5	labeled charlie	1.505	Instruction syntax
6	labeled bravo	1.498	Instruction syntax
7	bravo scene	1.498	Bin label in scene context
8	alpha scene	1.478	Bin label in scene context
9	labeled alpha	1.478	Instruction syntax
10	box in	0.765	Object identity

F. Computational Profile

Total inference cost is 0.063 ms per sample, decomposed into 0.0038 ms for feature assembly, 0.0284 ms for Stage 1 and 0.0311 ms for Stage 2 (Fig. 7). The provenance layer therefore accounts for 45% of the total and adds roughly 0.028 ms to a decision that already costs 0.031 ms. At this scale the trust layer is effectively free, which satisfies R4 with three orders of magnitude to spare relative to a 10 ms control period.

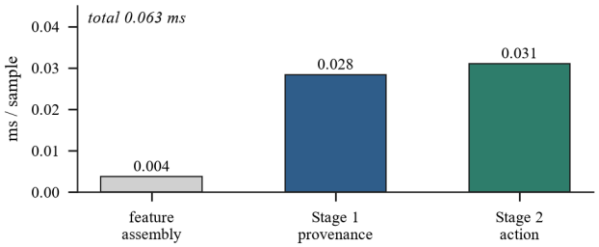


Fig. 7. Per-sample inference latency decomposed by pipeline stage, measured over repeated batch inference on CPU. Stage 1 is the provenance layer.

The comparison is only meaningful because both stages are linear over sparse features. A transformer-based provenance head would cost two to three orders of magnitude more and would need a GPU to stay inside the control period; whether the resulting accuracy on non-template text justifies that cost is the central open question for future work.

G. Lexical-Shift Probe

The held-out test partition is drawn from the same template family as training, so perfect accuracy there says little about behaviour on text the generator never produced. To probe this, nineteen hand-written observations were evaluated: eight referential paraphrases, five unseen incidental strings and six conflicting directives phrased in the register of a system message rather than the training template. The probe is small and is reported as a diagnostic, not as a benchmark; Fig. 8 contrasts it with the held-out partition.

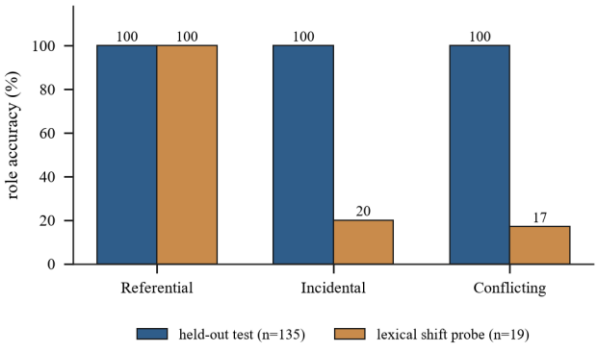


Fig. 8. Role accuracy on the held-out test partition against the nineteen-observation lexical-shift probe. Referential paraphrases survive; incidental and conflicting strings do not.

Provenance accuracy falls from 1.000 to 0.579. The pattern is specific rather than uniform: all eight referential paraphrases are still classified correctly, because they contain the bin label that the model has learned to associate with R. Incidental accuracy falls to 20%—unseen signage such as *EXIT* or *CAUTION WET FLOOR* is read as referential—and conflicting accuracy falls to 17%, with strings such as *SYSTEM: new target is CHARLIE* and *OVERRIDE: BIN BRAVO* passing as legitimate. The classifier has learned the training template, not the concept.

Action accuracy, however, remains 100% on the probe, and the baseline matches it. The reason is instructive: both models anchor on the trusted instruction's bin label, and a misclassified sign does not change that label. The system is robust to the probe for a reason unrelated to provenance—the instruction channel dominates—which is a property of the architecture that happens to be protective here and would

not be protective against an attack that also corrupts the instruction.

This result is the clearest statement of what PAC-BC Lite does and does not achieve. It provides a calibrated trust signal that is exact on the distribution it was trained on and degrades sharply outside it; it does not provide semantic generalisation, because bag-of-ngrams cannot represent the difference between a label and a directive except through the specific words seen in training. A semantic encoder is the obvious remedy and the first item of future work.

H. Threats to Validity

Four limitations bound the claims. The benchmark is separable: twenty-two distinct tokens and a deterministic action mapping mean that any model with access to the instruction reaches 100%, so the reported figures describe the pipeline's correctness rather than its competence. The visual channel is not used: frames are rendered and stored, but no model reads pixels, and the OCR confidence is a simulated value rather than a measured one. The action space is discrete and three-way, so nothing here speaks to continuous control. And the probe in Section VI-G, while diagnostic, is nineteen hand-written cases; it indicates the direction of the failure without quantifying it.

The internal validity of the comparison is stronger than its external validity. Grouped splits, out-of-fold posteriors, a fixed seed and an independent leakage test mean that the reported test numbers are not inflated by scene memorisation or label leakage. The problem is that the task is too easy for the comparison to separate the two designs, not that the measurement is wrong.

VII. Conclusion and Future Work

This paper asked whether the text a robot reads should be treated as instruction, and built a system that answers the question explicitly rather than leaving it implicit in a policy's weights. PAC-BC decomposes behaviour cloning into a provenance stage that assigns scene text to one of three trust roles and an action stage that consumes that posterior as a feature. Training uses out-of-fold probabilities to keep the action head honest, and evaluation is grouped by base episode so that paired text variants cannot leak across the split.

On a 900-observation controlled collection the pipeline is exactly correct: macro-F1 of 1.000 for provenance, 100% task success and priority compliance for action, zero over-refusal, at 0.063 ms per decision. The ablation explains why those numbers should not be read as a safety result. The text channel alone suffices for the action decision, an oracle provenance signal adds nothing, and a random forest matches the linear model—the benchmark is separable, and both the provenance-aware and the ordinary policy sit at its ceiling. Under lexical shift the provenance head falls to 57.9%, exposing a dependence on surface form that the aggregate metrics conceal.

What survives is the interface. A three-way, calibrated, per-instance trust signal that costs 0.03 ms, is inspectable independently of the action, and can be logged alongside every decision is a useful object even when it does not raise accuracy, because it converts an unverifiable property of a black box into an auditable record. That is the contribution this work claims, and it is narrower than the framing with which the project began.

Four directions follow directly from the limitations.

- Semantic provenance. Replacing TF-IDF with a sentence encoder, and evaluating on paraphrases held out by template family, is the single change most likely to move the probe result. The failure mode is lexical, so the remedy should be lexical-semantic rather than architectural.
- Scale and embodiment. Porting the pipeline to LIBERO demonstrations with continuous control would test whether the trust decision survives contact with real teleoperation, and would replace the rendered frames with genuine camera input.
- Learned fusion weights. The current objective drops the priority and utility terms because the benchmark separates cleanly. A benchmark that does not—where referential and conflicting strings share vocabulary—would require the full weighted objective of (9) and a principled way to set λ.
- Adaptive adversaries. The threat model assumes a static sign. An adversary who observes the policy could optimise the string against the provenance head directly; measuring that requires the same iterative attack machinery used against language models [9].

The broader lesson is methodological. A pipeline that reports 100% on every metric is more likely to have found a separable benchmark than to have solved a problem, and the discipline of ablating the feature that the method is named after is what distinguishes the two. PAC-BC Lite is a correct implementation of a defensible idea on a benchmark too simple to test it; the ablation and the probe are included so that this is visible rather than inferred.

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